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PI Name:	Shapiro, Irving M. Ph.D.		
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PI Email:	Irving.Shapiro@jefferson.edu	**Fax:**	FY 215-955-9159
PI Organization Type:	UNIVERSITY	**Phone:**	215-955-7217
Organization Name:	Thomas Jefferson University		
PI Address 1:	1015 Walnut St		
PI Address 2:			
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19107	**Congressional District:**	1
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Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
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Adaptation of bone cells to microgravity remains poor despite long term spaceflight experience. Histomorphometric studies of osteopenic bone point to microgravity affecting bone loss through a modest increase in resorption and a decrease in bone formation. In the context of this proposal, we propose to examine the impact of microgravity on terminal activities of osteoblasts, namely the induction of the apoptotic response. The major hypothesis to be tested is that microgravity promotes normal osteoblast development and induces bone cell apoptosis. There are three Specific Aims. The objective of Specific Aim 1 is to quantify the effects of microgravity on human bone cells. For this purpose the cells will be seeded onto hollow biodegradable microcarriers and maintained in rotating bioreactor that was designed by NASA for tissue engineering. Microgravity conditions will be imposed on the cells using interactive numerical modeling techniques that have been developed by our group. Development of the osteoblast culture will be assessed by measuring the activity of alkaline phosphatase, the formation of apatite and the expression of genes that characterize the

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bone cell phenotype. The second Specific Aim is directed at determining whether microgravity influences osteoblast function in terms of modulating bone cell proliferation and osteoblast death. Using a variety of techniques we will determine whether the induction of death is through activation of the apoptotic response. To understand the death pathway, we will then examine the impact of microgravity on expression of members of the Bcl-2 family of genes. We will test the hypothesis that microgravity causes activation of those genes that lead to apoptosis and a decrease in the expression of genes that protect cells from the apoptotic response. The last Specific Aim is to measure Ca2+ and Pi accumulation at the microcarrier under microgravity conditions and to determine whether the accumulated ions enhance osteoblast apoptosis. We willinvestigate whether microgravity causes an accumulation of ions that promote human bone cell apoptosis.Numerical modeling will compliment the experimental measurements by providing the surface rate accumulation of ions under conditions of simulated microgravity. Specific Aim 1. To quantify the effects of microgravity on human bone cells in the new microcarrier system in terms of (a) expression of the genes that are characteristic of bone (Osf-2, osteocalcin, osteopontin and type I collagen); (b) the presence of biological apatite (c) the activity of alkaline phosphatase
Hypothesis 1: microgravity modulates the expression of major osteogenic genes, the synthesis of the mineralized phase of bone and the maturation of the bone cell phenotype.

Numerical modeling: will provide information on hydrodynamic variables that govern shear stress at the cell surface under microgravity conditions; it will be used to generate data on the optimum location of cells in the rotating wall vessel with adequate information on nutrient levels and pO2 .

Specific Aim 2. To examine evaluate the effects of microgravity on osteoblast function in terms of bone cell proliferation and cell death. To ascertain whether microgravity induces human bone cell apoptosis.

Hypothesis 2: microgravity induces bone cell death by enhancing the apoptotic process.

Numerical modeling: same as Aim 1.

Specific Aim 3. To measure Ca2+ and Pi accumulation at the microcarrier under microgravity conditions and to determine whether the accumulated ions enhance osteoblast apoptosis.

Hypothesis 3: To investigate whether microgravity causes an accumulation of ions that promote human bone cell apoptosis.

Numerical modeling: numerical modeling will compliment the experimental measurements by providing the surface rate accumulation of ions under conditions of simulated microgravity. The other aspects of numerical modeling remain the same as Aim 1. Knowledge of the adaptation of the human bone cells to microgravity remains poor despite long term space flight experience and the availibilty and accurate techniques for bone mass measurements. Microgravity has been noted to modify the function of bone cells and disturb metabolism. Cellular changes are reported to oocur early and if prolonged result in osteopenia. Bone loss is thought to be due to a modest loss of bone through resorption and a decrease in bone formation. Some studies suggest that reveresibilty of bone loss is unlikely after six months of immobilization or weightlessness. For this reason, prolonged weightloss of astronauts may present problems on return to Earth with increased risk of fractures and premature osteoporosis. Thus, there is a great need to gain to a fundamental understanding of the effects of microgravity on human bone cell function. In the context of this investigation, we are examining the impact of microgravity on terminal activities of osteoblasts, namely the induction of the apoptotic response. Our major hypothesis is that microgravity promotes bone cell apoptosis.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

BIOCHEMISTRY AND CELL BIOLOGY STUDIES
Effects of Vector Averaged Gravity on Osteoblasts and Osteoclasts

Using our new system for generating vector averaged gravity, MC3T3 osteoblast-like cells were cultured in alginate beads for periods of time ranging from 1-14 days. We evaluated osteogenic gene expression by RT-PCR analysis. All of the cells expressed RUNX2 and osteocalcin (OCN). No significant differences were observed between cells in maintained in microgravity and normal gravity. This finding lends support to our contention that vector averaged gravity does not activate the apoptotic cascade. The finding also confirms that our new system maintains the osteoblast-like phenotype.

Although the cells did not undergo spontaneous apoptosis in vector-averaged gravity, it was hypothesized that the cells would be compromised. Thus, the cells would die when challenged with apoptosis-inducing agents. To explore this hypothesis, the cells were challenged with staurosporine and sodium nitroprusside after 24 h culture in the HARV. There was no detectable difference in the viability of cells maintained in microgravity and unit gravity. When treated with the apoptogen staurosporine and sodium nitroprusside there was a dose dependent reduction in viability of the treated cells. The change in viability was independent of exposure to vector averaged gravity suggesting that the cells were not sensitized to these apoptogens by simulated microgravity.

We increased the effective gravity by subjecting cells to hypergravity using a modified centrifugation technique. Osteoblasts in alginate carriers were cultured in a centrifuge (Centra CL3R, IEC). The centrifuge was coupled with an incubator to maintain 37C and 5% CO2 and custom programmed to operate at a minimum of 200 rpm. By culturing cells at different radial positions, two RCF values were evaluated in a single experiment (3 g and 6 g). While no changes in viability were observed due to vector averaged gravity, preliminary results using the centrifuge system suggest the hypergravity decreases the viability of osteoblasts.

Use Of The New Modified HARV System To Study Bone Resorption A further modification of the carrier system is the incorporation of a bone into the alginate carrier. Intact adult mouse distal phalanges were subjected to simulated microgravity in the HARV for 5 days. The new system has the advantage that we can now evaluate the mineralization status of intact bones subjected to microgravity. To evaluate and compare the effect of microgravity and unit gravity on osteoclast formation, the expression of RANK and cathepsin K were determined by mRNA in situ hybridization. An increase in RANK expression was observed after 5 days of exposure to microgravity. The level of RANK expression was elevated above values observed in bones exposed to unit gravity. Similarly, the level of cathepsin K expression was

Task Progress:	elevated in microgravity as compared to unit gravity. In addition, the number of osteoclasts present in bones that had been exposed to microgravity was determined by enzyme histochemistry. There was an increase in the number of cells that expressed TRAP in bones exposed to microgravity as compared with bones in normal gravity. The intensity of the stained cells was also increased. NUMERICAL STUDIES ON PHOSPHATE KINETICS AND HARV DYNAMICS Numerical Modeling Of Phosphate Ion Transfer In The Harv: Application Of Michaelis- Menten Kinetics For Phosphate Ion Consumption By Napi Co-Transporter System We have modeled the transport and accumulation of the phosphate ions in a basic multicellular unit consisting of osteoclasts and osteoblasts situated adjacent to each other in straining flows. Two such flows are considered; One corresponding to shear levels representative of trabecular bone conditions at normal gravity, and the other corresponding to shear level that is representative of simulated microgravity conditions prevailing in a NASA designed Rotating Wall Bioreactor (RWV) culture. The consumption of Pi by osteoblasts is assumed to follow Michaelis-Menten (MM) kinetics aided by NaPi co-transporter system. MM kinetics views the NaPi co-transporter as a system for transporting extracellular Pi into the osteoblast. Our results show, for the condition investigated here, the net accumulation of the phosphate ions at the osteoblast surface at simulated microgravity conditions is higher by as much as a factor of 4. Such increased accumulation may lead to enhanced apoptosis and may help explain the increased bone loss observed in microgravity. Numerical Modeling Of Particulate Motion In The HARV: Relevance Of The Rotating Frame Of Reference Detailed numerical simulation results for the three-dimensional motion of a single or many spherical rigid particles in a Newtonian-fluid filled cylindrical vessel rotating about its horizontal axis are provided. The ranges of low and intermediate particle Reynolds number have been investigated. We have developed a new DNS based on the Arbitrary Lagrangian-Eulerian (ALE) moving grid unstructured finite element scheme and is described in a frame of reference with its origin coinciding with the center of the cylinder and rotating with the fluid (a rotating frame of reference). The newly developed model is noted to provide an accurate description of the flow and of the particulate behavior. Effecting the solution in the rotational frame of reference is seen to be computationally very fast and efficient for this problem compared to conventional computations in an inertial frame of reference. Several novel features in regard to two particle interactions in rotating flows have been revealed by this detailed numerical study. EMPIRICAL DATA ON ALGINATE CARRIER-HARV SYSTEM A novel apparatus, based on a previously reported method, was used to encapsulate osteoblast-like cells in alginate carriers. Carriers containing MC3T3 cells were cultured together in a 50 ml re-usable HARV (Synthecon). Prior to loading the HARV with carriers, we used Ficoll 400 (Sigma) to increase the density of the culture medium. The buoyancy of the carriers minimized collisions between the alginate particles and the walls of the HARV. We evaluated the trajectory of the carriers in the HARV by video and video microscopy using a camera (XL1S, Canon) with a macro lens. The camera tracked the motion and orientation of carriers in the HARV. The vector-averaged gravity experienced by the cells was calculated using the cosine of the angular position of the carriers. These results correlated very well with the predictions made by numerical simulations. Over the course of one revolution, the cells in polar and nonpolar carriers experience time-averaged gravitational accelerations of 0.993g (+/-0.005), and 0.088g (+/-0.084) respectively.
Bibliography Type:	Description: (Last Updated: 05/15/2006)
Abstracts for Journals and Proceedings	K.M.Krishnan, H.H. Hu "Lift Force on a Sphere Moving in a Rotating Cylinder" APS/DFD Annual Meeting, 2002 1 , Jan-2003
Abstracts for Journals and Proceedings	K.M.Krishnan, P. Ayyaswamy, H.H. Hu ", Hydronamic Interaction of Two Particles Suspended in a Horizontally Rotating Cylinder. " APS/DFD Annual Meeting, 2003 2 , Jan-2003
Abstracts for Journals and Proceedings	K.M.Krishnan, P. Ayyaswamy, M. Bucaro, M. Risbud, I.M. Shapiro, C.S. Adams. "Modelling of Phosphate Ion Transfer to the Surface of Osteoblasts in a Rotating Wall Vessel" NASA Cell Science Conference, 2003 3 , Mar-2003
Abstracts for Journals and Proceedings	M. Bucaro, I.M. Shapiro, C.S. Adams, K. Mukundkrishnan, P Ayyaswamy, M V. Risbud "Modeled Microgravity Increases Sensitivity of Osteoblasts to Apoptogens" NASA Cell Science Conference, 2003 4 , May-2003
Abstracts for Journals and Proceedings	I.M . Shapiro, M. Bucaro, C. S Adams, M. Steinbeck, P.S. Ayyaswamy, K. Mukundkrishnan and M. V. Risbud. "Bone Cell Survival in Microgravity: Evidence that Modeled Microgravity Increases Osteoblast Sensitivity to Apoptogens" Microgravity Transport Processes in Fluid, Thermal, Biological and Material Sciences III, ECI, Davos, Switzerland, September, 2003 5 , Jan-2004
Abstracts for Journals and Proceedings	M. Bucaro, A.M. Zahm, M. Steinbeck, P.S. Ayyaswamy, K. Mukundkrishnan and M. V. Risbud, I.M . Shapiro, C. S Adams "A Novel Method For Evaluating Osteoblast Apoptosis in Simulated Microgravity." NASA Cell Science Conference, 2004. NASA Cell Science Conference, April 2004. , Apr-2004
Abstracts for Journals and Proceedings	M. Bucaro, A.M. Zahm, M. Steinbeck, P.S. Ayyaswamy, K. Mukundkrishnan and M. V. Risbud, I.M . Shapiro, C. S Adams "A Novel Alginate Carrier System to Evaluate Osteoblast Apoptosis in Vector-averaged Gravity using Rotating Wall Vessels" Meeting of Bone Mineral Research Society, August 2004. J. Bone Min. Res. 19: 392. 2004. , Aug-2004
Abstracts for Journals and Proceedings	H.H. Hu, P.N.Swaminathan, K.M.Krishnan. "Simulations of an Ellipsoid Settling in Cylindrical Tube at Low and Intermediate Reynolds Numbers." APS/DFD Annual Meeting, 2004. APS/DFD Annual Meeting, November 2004 . , Nov-2004

Articles in Peer-reviewed Journals	M.A. Bucaro, J. Fertala, C.S. Adams, M. Steinbeck, P. Ayyaswamy, K. Mukundakrishnan, I.M. Shapiro, M.V. Risbud "Bone Cell Survival in Microgravity: Evidence that Modeled Microgravity Increases Osteoblast Sensitivity to Apoptogens." Ann. N.Y. Acad. Sci. 1027: 1–10, 2004, Jan-2004 . , Jan-2004
Articles in Peer-reviewed Journals	Mukundakrishnan K, P.S. Ayyaswamy, M. Risbud, H.H. Hu, I.M. Shapiro. "Modelling of Phosphate Ion Transfer to the Surface of Osteoblasts under Normal and Simulated Microgravity Conditions." Ann. N.Y. Acad. Sci.1027: 10–14, 2004, Aug-2004 , Aug-2004